

Central MIS — Local Setup

NGA Central MIS · run it on your own laptop · about 20 minutes

Central MIS is the core school system *and* the sign-in provider for every other NGA app. It is a React app (frontend/, http://localhost:5173) and an Express API (backend/, http://localhost:5001) on a MySQL database. Because it is the identity provider, it has real user accounts of its own — you log in with a local super-admin, no SSO secret needed.

Before you start

Install **Node.js 20 or newer**, **Git**, and **MySQL 8** (nodejs.org · git-scm.com · dev.mysql.com/downloads/installer — choose “Server only”). Note the MySQL **root password** you set; you need it in step 3.

1 · Get the code

```
git clone https://github.com/niyongaboemmy/nga_central_mis.git
cd nga_central_mis
npm run install-all
```

Installs the root, backend and frontend. Takes a few minutes the first time.

2 · Create your settings files

```
Copy-Item backend\.env.example backend\.env
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```

On macOS or Linux use `cp backend/.env.example backend/.env` and the same for frontend.

3 · Fill in backend/.env

Open backend/.env and set one value. Everything else works as-is locally; frontend/.env needs no changes.

Setting	Value
DB_PASSWORD	your MySQL root password (leave empty if you set none)

4 · Create the database

```
cd backend
npm run db:setup
cd ..
```

Builds the ngarw_mis database from the snapshot in the repo, brings the schema up to date, and sets the super-admin password. It refuses to overwrite an existing database — add `--force` to rebuild one.

5 · Run it

```
npm run dev
```

Backend and frontend start together and reload as you edit. Open **http://localhost:5173** and log in:

Username	superadmin
Password	Admin@1234
Code	login is two-step; in development the code is shown on the login page, no email is sent

Stop with Ctrl+C.

6 · Working on a feature

Pushing to main deploys to production. Always work on a branch and open a Pull Request:

```
git checkout -b feature/short-description
# make and test your changes, then
git push -u origin feature/short-description
```

If something goes wrong

You see	Do this
<i>Access denied for user 'root'</i>	DB_PASSWORD in backend/ .env is wrong.
<i>ECONNREFUSED 127.0.0.1:3306</i>	MySQL is not running. Start the "MySQL80" service (Windows Services) or <code>brew services start mysql</code> .
<i>Database already has N tables</i>	You ran <code>db:setup</code> before. Use <code>npm run db:setup -- --force</code> to rebuild.
<i>Unknown column ... after pulling new code</i>	A teammate added a migration. Apply it: <code>npm run migrate NNN_name.sql</code> in backend/, or rebuild with <code>db:setup -- --force</code> .
<i>Port 5001 / 5173 is already in use</i>	Stop the other process (<code>netstat -ano findstr :5001</code>) and start again.
Want a clean database	<code>cd backend</code> then <code>npm run db:setup -- --force</code> .

Checks: `curl http://localhost:5001/health` · `npm test` in backend/ · `npm run lint` in frontend/